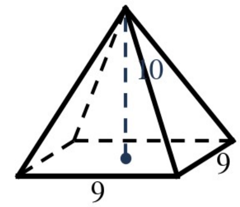
Speak **L**ike **A M**athematician:



slant height of a cone:

the one-third rule:

Pyramids and Cones

Pyramid: V = (1/3) B h. Cone: V = (1/3) π ______ h.

Cone slant height: $\ell^2 = r^2 +$ ______ (a right triangle inside the cone).

Cone surface area: SA = $\pi r^2 + \pi r$ ______ .

Example 1 – Square Pyramid

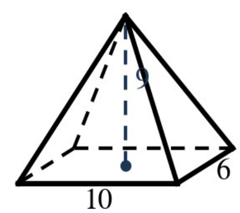
The pyramid at the top right has a 9-by-9 base and height 10.

a) Find its volume.

Example 2 – Rectangular Pyramid

The pyramid at the right has a 10-by-6 base and height 9.

a) Find its volume.



Example 3 – Working Backward

a) A pyramid has volume 64 and height 12. Find the base area, and the side of its square base.

Example 4 – Triangular Pyramid

a) The base is a triangle with base 5 and height 12; the pyramid is 8 tall. Find V.

Example 5 – A Big One

a) A square pyramid has base side 15 and height 20. Find its volume.

b) How does that compare to the box (prism) with the same base and height?

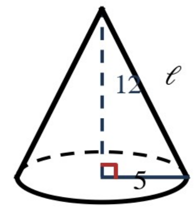
Example 6 – Cone Volume and Surface Area

The cone at the right has radius 5 and height 12.

a) Find the slant height ℓ.

b) Find the volume (exact and nearest tenth).

c) Find the surface area.



Example 7 – Working Backward

a) A cone with radius 4 has volume 48π. Find its height.

Example 8 – A Curious Cone

The cone at the right has radius 6, height 8, slant height 10.

a) Compute BOTH the volume and the surface area. What do you notice?